

V6.2 · Semantic Hierarchy Architecture

Soft, multi-scale, graph-structured interpretation above a frozen foundation

Objective	Replace the single hard 13-theme layer of V6 with a soft, multi-resolution hierarchy that preserves spectroscopic information and represents genuine chemical overlap.
Method	Two non-negative linear maps above the frozen atlas: M (component→motif, frozen from V6) and S (motif→theme, soft and row-stochastic, new in V6.2).
Frozen	Atlas 09ed804a40836f4a05a91ba10900cded; the V6 MSS map M is an input, never rewritten.
Reproduce	results/v6_rebuild/code/p10 → p11 → p12 → p13

Philosophy

GAIRA is a biochemical reasoning engine, not a molecular classifier. V6.2 optimises **maximum biochemical abstraction subject to minimum information loss**. Priority order: interpretability, information retention, recoverability, and only then analyte-level accuracy — a modest accuracy loss is an acceptable price for a representation that transfers.

The chain

$\text{coord}(x) = \text{NNLS}(\text{preprocess}(x), H)$ 24 non-negative shares, frozen

$\text{mss}(x) = M^T \text{coord}(x)$ 17 spectroscopic motifs, frozen from V6

$\text{theme}(x) = S^T \text{mss}(x)$ K soft chemical themes, NEW

$S \geq 0$, $\sum_t S[m,t] = 1$, sparse

The V6.2 semantic hierarchy — multi-scale, soft, non-circular

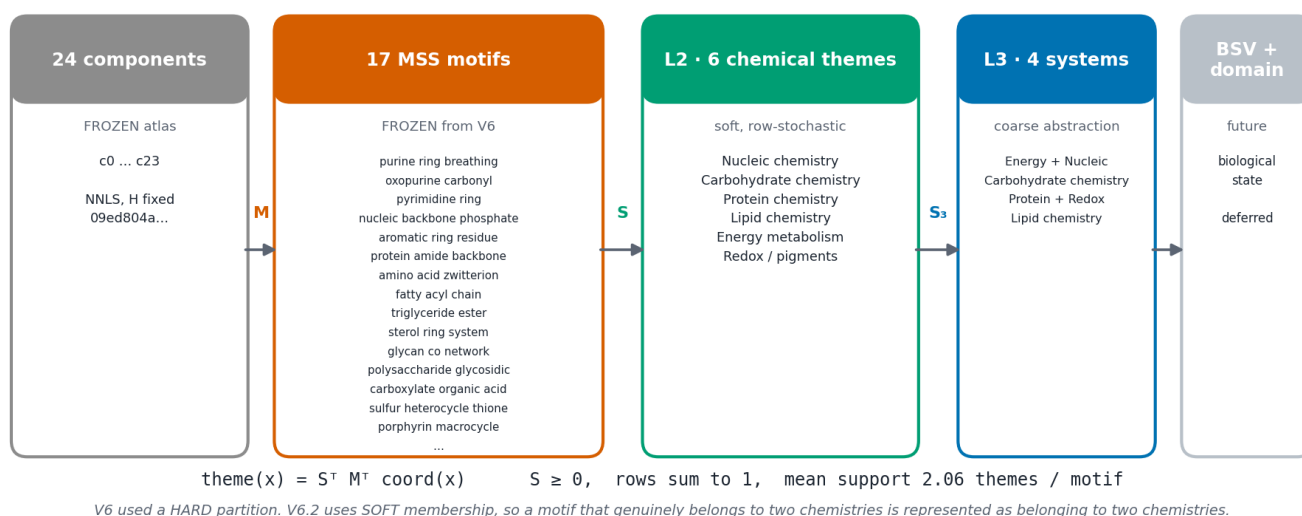


Figure 1 — The V6.2 hierarchy at three semantic resolutions.

The three resolutions

level	K	chemically admissible	how derived
L1 fine	17	yes	the MSS motifs themselves
L2 medium	6	yes	K=8 hybrid clustering was chemically inadmissible; fell back to the ch
L3 coarse	4	no	agglomerated from the L2 theme centroids to K=4

L2 chemical theme	member MSS motifs
Nucleic chemistry	oxopurine_carbonyl, purine_ring_breathing, pyrimidine_ring
Carbohydrate chemistry	glycan_co_network, nucleic_backbone_phosphate, polysaccharide_glycosidic
Protein chemistry	amino_acid_zwitterion, aromatic_ring_residue, protein_amide_backbone
Lipid chemistry	fatty_acyl_chain, sterol_ring_system, triglyceride_ester
Energy metabolism	carboxylate_organic_acid, sulfur_heterocycle_thione
Redox / pigments	carotenoid_polyene, flavin_redox_cofactor, porphyrin_macrocycle

L3 biochemical system	member MSS motifs
Energy + Nucleic	carboxylate_organic_acid, oxopurine_carbonyl, purine_ring_breathing, pyrimidine_ring, sulfur_heterocycle_thione
Carbohydrate chemistry	glycan_co_network, nucleic_backbone_phosphate, polysaccharide_glycosidic
Protein + Redox	amino_acid_zwitterion, aromatic_ring_residue, carotenoid_polyene, flavin_redox_cofactor, porphyrin_macrocycle, protein_amide_backbone
Lipid chemistry	fatty_acyl_chain, sterol_ring_system, triglyceride_ester

An honest negative result. The K=8 grouping produced by the hybrid motif distance was **chemically inadmissible** (it merged motifs from unrelated superclasses), so L2 falls back to the chemical-superclass grouping at **K=6**, not the 8 suggested in the brief. The data did not support 8 derived themes; the number was not forced.

Interpretation

The architectural change is that a motif is no longer owned by one theme. 11 of 17 motifs carry membership in more than one chemical theme, with a mean of 2.06 themes per motif. That is not noise: it is the spectroscopic reality that amide III and the saccharide C-O network occupy the same region.

Limitations

- L2 is the chemical-superclass grouping, not a data-derived one — the derived K=8 clustering failed the admissibility rule.
- L3 is chemically inadmissible by the strict L2 rule, which is expected at a coarse level (it merges superclasses by design) but means the rule does not constrain it.
- The soft-membership temperature is a chosen parameter, selected by a stated sparsity criterion (§ Soft Theme Membership) rather than derived from first principles.

Recommendations

- Treat L2 as the operational level and L1 (motifs) as the evidence level; L3 is for communication, not inference.

- Revisit the derived K=8 grouping once the corpus contains pure porphyrin, flavin and sterol references — the inadmissibility traces to those under-grounded motifs.

Atlas 09ed804a40836f4a05a91ba10900cded verified unchanged. All values recomputed from results/v6_rebuild/.